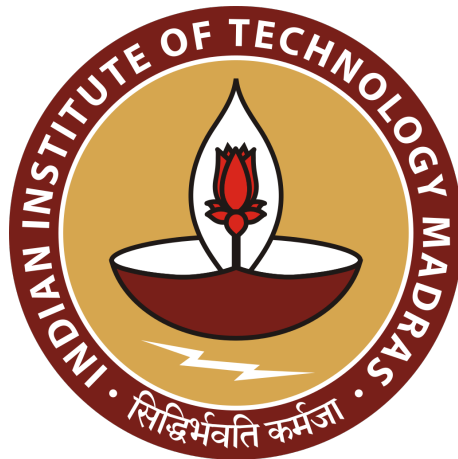


Lab Assignment - 5

Database Querying, Multiple Sequence Alignment



BT3040 - Bioinformatics

Even Semester 2026

Course Instructor: Professor Michael Gromiha

K.Varunkumar | BE23B016

Department of Biotechnology

Indian Institute of Technology, Madras

Question 1

BLASTP was used with the “nr” database, the following protein sequences are found to be similar to the given sequence:

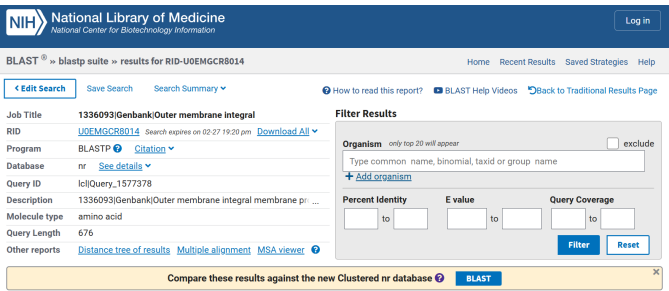


Figure 1.1.1: BLASTP using "nr" database

Sequences producing significant alignments									
select all 100 sequences selected									
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per ident	Acc. Len	Accession	
type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1387	1387	100%	0.0	100.00%	676	WP_004150386.1	
type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1385	1385	100%	0.0	99.85%	676	WP_168385178.1	
type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1385	1385	100%	0.0	99.85%	676	WP_168421624.1	
type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1383	1383	100%	0.0	99.85%	676	WP_004150705.1	
type III secretion system outer membrane ring subunit SctC [Erwinia amylovora]	Erwinia amylovora	1371	1371	100%	0.0	98.97%	677	WP_004150436.1	
Type III secretion system outer membrane pore HscC [Erwinia amylovora ATCC BAA-2158]	Erwinia amylovora	1369	1369	100%	0.0	98.82%	677	C80070.967.1	
tracheal type III secretion system Htrach secretion translocation pathway htrC protein [Erwinia sp. Eip617]	Erwinia sp. Eip617	1347	1347	100%	0.0	96.89%	676	AB211387.1	
type III secretion system outer membrane ring subunit SctC [Erwinia carotovora]	Erwinia carotovora	1343	1343	100%	0.0	96.75%	676	WP_252681678.1	
tracC [Erwinia carotovora]	Erwinia carotovora	1339	1339	100%	0.0	96.45%	676	WP_012680302.1	
type III secretion system outer membrane ring subunit SctC [Erwinia carotovora]	Erwinia carotovora	1337	1337	100%	0.0	96.30%	676	AB230758.2	
type III secretion system outer membrane ring subunit SctC [Erwinia carotovora]	Erwinia carotovora	1289	1289	100%	0.0	92.40%	676	WP_023633761.1	
Erwinia carotovora	Erwinia carotovora	1242	1242	97%	0.0	93.62%	676	WP_012440286.1	
Erwinia carotovora	Erwinia carotovora	1211	1211	100%	0.0	88.67%	677	WP_124232187.1	
type III secretion system outer membrane ring subunit SctC [Erwinia trachealis]	Erwinia trachealis	1184	1184	100%	0.0	83.85%	674	WP_023452988.1	
MULTISPECIES: type III secretion system outer membrane ring subunit SctC [Phenol]	Phenol	1184	1184	100%	0.0	84.09%	679	WP_275222444.1	
type III secretion system outer membrane ring subunit SctC [Erwinia trachealis]	Erwinia trachealis	1183	1183	100%	0.0	83.70%	674	WP_015101020.1	
type III secretion system outer membrane ring subunit SctC [Phenol sp. AS142]	Phenol sp. AS	1181	1181	100%	0.0	84.09%	679	WP_337013830.1	

Figure 1.1.2: List of significant alignments

Range 1: 1 to 676 GenPept Graphics									
▼ Next Match ▲ Previous Match									
Score	Expect	Method	Identities		Positives		Gaps		
1387 bits(3589)	0.0	Compositional matrix adjust.	676/676(100%)		676/676(100%)		0/676(0%)		
Query 1	MVEKRELRCRLLGALLMLCATLPAGAQTADWKEQSYAYSADRTPLSTVLQDFADGHVSVD								60
Sbjct 1	MVEKRELRCRLLGALLMLCATLPAGAQTADWKEQSYAYSADRTPLSTVLQDFADGHVSVD								60
Query 61	LHLGNVEDTEVTAKIRAENASAFDLRLALEHHFQWVFNNTLYVSPQDEQSSERLEISPD								120
Sbjct 61	LHLGNVEDTEVTAKIRAENASAFDLRLALEHHFQWVFNNTLYVSPQDEQSSERLEISPD								120
Query 121	AAPDIKQALSIGILLDPDFGWGELPDDGVVLVTGPPQYLELVKRFSEQRKKEDRRKVM								180
Sbjct 121	AAPDIKQALSIGILLDPDFGWGELPDDGVVLVTGPPQYLELVKRFSEQRKKEDRRKVM								180
Query 181	FPLRYASVADRTIHYRDQTVVIGVATMLNELMNGKRAAPASASGIDSTPGGPDNTSM								240
Sbjct 181	FPLRYASVADRTIHYRDQTVVIGVATMLNELMNGKRAAPASASGIDSTPGGPDNTSM								240
Query 241	NTQTLRLSRLSSRNKTSNRAGGRDNEIEDVSGRISADVRNNALLIRDDDKRHDEYSQ								300
Sbjct 241	NTQTLRLSRLSSRNKTSNRAGGRDNEIEDVSGRISADVRNNALLIRDDDKRHDEYSQ								300
Query 301	IDVPQNLVEIDAVILDIRTALNRLLEANWQATLGGVTGGSSLMMSGSGTLFVSDFKR								360
Sbjct 301	IDVPQNLVEIDAVILDIRTALNRLLEANWQATLGGVTGGSSLMMSGSGTLFVSDFKR								360
Query 361	IQALEGEGTASIVANPSVLTLNQPAVIDFSQTAYITATGERVADIQPVTAGTSLQVTP								420
Sbjct 361	IQALEGEGTASIVANPSVLTLNQPAVIDFSQTAYITATGERVADIQPVTAGTSLQVTP								420
Query 421	AVGNEGHSIIQLMIDIEDGHVQTNGDQATGVKRGTVSTQALISENRLVLGGFHVEESA								480
Sbjct 421	AVGNEGHSIIQLMIDIEDGHVQTNGDQATGVKRGTVSTQALISENRLVLGGFHVEESA								480
Query 481	DRDRRIPLLDIPWLGQLFSSKRHEISQRQLFILTPRLIGDQTDTPTRYVTADNRQLSD								540
Sbjct 481	DRDRRIPLLDIPWLGQLFSSKRHEISQRQLFILTPRLIGDQTDTPTRYVTADNRQLSD								540
Query 541	AMGRVERRHSSVNQHDVVENALRDLAEGQSPAGFQPQTSGTRLSEVCRSTPALLFESTRG								600
Sbjct 541	AMGRVERRHSSVNQHDVVENALRDLAEGQSPAGFQPQTSGTRLSEVCRSTPALLFESTRG								600
Query 601	QWYSSSTNGVQLSVGVVRNTSSKPLRFDEANCASKRTLAVAVWPHSALAPGESAEVYL								660
Sbjct 601	QWYSSSTNGVQLSVGVVRNTSSKPLRFDEANCASKRTLAVAVWPHSALAPGESAEVYL								660
Query 661	DPSRVLHASRESLLNR 676								
Sbjct 661	DPSRVLHASRESLLNR 676								

Figure 1.1.3: Highest scoring sequence

- All 100 similar sequences have an E value of 0
- Only 1 sequence has 100% identity (type III secretion system outer membrane ring subunit SctC [Erwinia amylovora])
- The above sequence also holds the highest alignment score at 1387

BLASTP was used with the “swissprot” database, the following protein sequences are found to be similar to the given sequence:

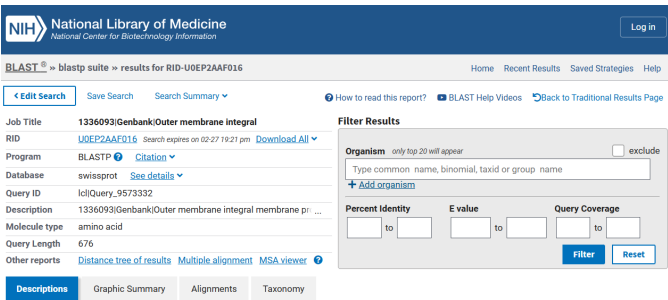


Figure 1.2.1: BLASTP using "swissprot" database

Sequences producing significant alignments										
Download Select columns Show 100										
GenPept Graphics Distance tree of results Multiple alignment MSA viewer										
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	566	566	99%	0.0	44.16%	701	Q01723.2		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	251	251	72%	3e-73	31.25%	607	Q01723.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	246	246	76%	1e-71	36.52%	607	Q01723.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	211	211	75%	5e-69	28.87%	560	Q02095.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	175	175	70%	6e-46	27.63%	555	Q05641.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	175	175	70%	6e-46	27.63%	555	Q05641.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	158	158	73%	2e-40	26.13%	407	Q02095.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	132	132	37%	6e-31	31.25%	407	Q02095.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	104	104	37%	1e-22	30.00%	412	Q34749.2		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	92.4	92.4	37%	3e-18	28.97%	678	Q31780.2		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	92.0	92.0	42%	5e-18	28.95%	712	Q31780.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	90.1	90.1	37%	2e-17	28.42%	678	Q31780.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	89.0	89.0	42%	4e-17	27.66%	710	Q01565.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	86.7	86.7	84%	2e-16	22.80%	650	Q31781.2		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	79.7	79.7	37%	2e-14	26.41%	445	Q31782.2		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	79.0	79.0	76%	6e-14	23.45%	660	Q15641.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	76.3	76.3	41%	4e-13	22.63%	714	Q31782.2		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	73.2	73.2	42%	4e-12	23.97%	701	Q02095.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	72.8	72.8	42%	5e-12	23.97%	709	Q02095.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	72.8	72.8	42%	5e-12	23.97%	707	Q02095.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	71.6	71.6	42%	1e-11	23.91%	712	Q02095.1		
✓ Biachne_FullType_3 secretion system secretin: Short:1335 secretin: ABName_FullType3secretin: Pseudomonas sy...	Pseudomonas sy...	68.9	68.9	28%	7e-11	27.62%	686	F3P386.1		

Figure 1.2.2: List of significant alignments

Range 1: 1 to 691										GenPept	Graphics	▼ Next Match	▲ Previous Match
Score	Expect	Method	Identities				Positives		Gaps				
566 bits(1460)	0.0	Compositional matrix adjust.	310/702(44%)				444/702(63%)		47/702(6%)				
Query	7	LRCRLGALLMLCATLPA-GAQTADWKEQSYAYSADRTPLSTVLQDFADGHSVDLHLGN									65		
Sbjct	1	+R L+ L+L PA A TP WK +YAY A +T L+T L DFA + L +									60		
MRKALMWLPLLLIGLSPATWAVTPEAKWHTAYAYDARQTELETALADFAKEFGMALDMPP													
Query	66	VEDTEVTAKIRAENASAFDLRLALEHHFQWFVYNNNTLYVSPQDEQSSERLEISPDAAPI									125		
Sbjct	61	+ + +IRA++ FLDRLE +HFQWFVYN+TLYVSP E+S R+E+S DA D+									119		
IPGV-LDDRIRASQPEEFLDRLGQEHYHFWFYNDTLYVSPSEHTSARIEVSSDAVDL													
Query	126	KQALSGIGLLDPRFGWGLPDDGVVLVTGPPQYELVKRFSEQR--KKEDRRKVMFTPL									183		
Sbjct	120	+ AL+ +GLLD RFGWG LP++GWL V GP +Y+ELV+ +S++ E +K D++ V+ FPL									179		
QTALT DVGLLDKRFGWGLPNEGVLVVRGPAKYVELVRDYSKKVEAPEKGDKQDVIVFPL													
Query	184	RYASVADRTIHYRDQTVVPIGVATMLNLMNGKRAAPASAGDS-----									228		
Sbjct	180	+YAS ADRTI YRDQ +V+ GVA++L +L++ R+ S +G+D									238		
KYASAADRTIYRDQQLVWGVASILQDLLD-TRSHGSGINGMDLLGRGGRGNLAGGGS													
Query	229	--TPGGP-----DTNMMQNTQTLL--SRLSSRNKTSNRAGGRDNEIEDVSGRISADVR									278		
Sbjct	239	TP P DTN++ Q +L +++ +R+GGR N R++ADVR									292		
PDTPLSPMSSSGLDNTALEQGLDQVLYHGGGKTSKSGKSRSGGRANI-----RVTADVR													
Query	279	NNALLIRDDDKRHDEYSQLIKIDVPPQNLVEIDAVILDIRTALNRLANWQATLGGVTG									338		
Sbjct	293	NNA+LI D R Y +LI +DV +NL+EIDAVILDIR L L + W G V G									352		
NNA+LIYDLPSRKAMEYKLIKELDVSRNLEIDAVILDIRNELAELSSRWNFNAGSVNG													
Query	339	GSSLSMSG--SGTLFVSDFKRFADIAQLEGGTASIVANPSVLTLENQPAVIDFSQTAYI									396		
Sbjct	353	G+++ S TLF+ + +F A++ ALEG G+AS++ NPS+LTLENQPAVIDFS+T Y+									412		
GANMFDAQTSSSTLFIQNAKGFAAELHALEGNASASVIGNPSILTLENQPAVIDFSRTEYL													
Query	397	TATGERVADIQPVTAGTSLQVTPRAVGNHSSIIQLMIDIEDGHVQTN--GDGQATGVKR									454		
Sbjct	413	TAT ERVA+I+P+TAGTSLQVTPR++ +G +QL++DIEDG + + D Q + V++									471		
TATSERVANIEPTAGTSLQVTPRSLDHDGKPQVQLIVDIEDGQIDISDINDTQPS-VRK													
Query	455	GTVSTQALISENRLVLGGFHVEESADRRRIPLLDIPWLQGLFSSKRHEISQRQLRF									513		
Sbjct	472	G VSTQA+I+E+ +LV+GGFH E+ D+ ++PLLGDIP++G+ LF S+ E+SQR+RLF									531		
GNVSTQAVIAEHGSLVIGGFHGLEANDKVKHVPVLLGDIPIYIGKLLFQSRSELSQRERLF													
Query	514	ILTPRLIGDQTDPTTRYVTADNRQQLSDAMGRVRRH--SSVNQHDVVENALRDLAEGQSP									571		
Sbjct	532	ILTPRLIGDQ +P RYV N + D M R++ R+ ++ ++ +G +P									591		
ILTPRLIGDQVNPARYVQNGNPHDQDKMKIKERRDGELPTRGDIQVFTQMVDGAAP													
Query	572	AGFQP-QTSGTRLSEVCRSTPALLFESTRGQWYSSSTNGVQLSVGVVRRNTSSKPLRFDEA									630		
Sbjct	592	G +T +C L + R QWY+ GV +V V RN + KP+R DE+									649		
EGMHGDETLPFETDSLCPGQGLSLDGQSRQWYARKDWGV--AVVVARNTDKPVRIDES													
Query	631	NCASKRTLAVAVWPHSALAPGESAEVYLAMPSPRVLHASRES									672		
Sbjct	650	C + + VA WPH+ L PGE +EVY+A+ ++ ++ES									691		
RCGGRWVIGVAAWPHAWLQPGEESEVYIAVRQPQISKMAKES													

Figure 1.2.3: Highest scoring sequence

- All 33 similar sequences have an E value very close 0
- No sequence has 100% identity in this case (44.16% being the highest)
- The sequence with the greatest alignment is type III secretion system secretin [Pseudomonas syringae pv. syringae], with a score of 566

Question 2

The algorithmic parameters used for the searches performed in Q1 are given below:
(NOTE: The default parameters were used in both cases.)

The screenshot shows the 'Algorithmic parameters' page for BLAST. It is divided into three main sections: General Parameters, Scoring Parameters, and Filters and Masking. A note at the top states: 'Note: Parameter values that differ from the default are highlighted in yellow and marked with a sign'. A 'Restore default search parameters' button is in the top right.

- General Parameters:**
 - Max target sequences: 100 (dropdown)
 - Short queries: ☒ Automatically adjust parameters for short input sequences
 - Expect threshold: 0.05 (text input)
 - Word size: 5 (dropdown)
 - Max matches in a query range: 0 (text input)
- Scoring Parameters:**
 - Matrix: BLOSUM62 (dropdown)
 - Gap Costs: Existence: 11 Extension: 1 (dropdown)
 - Compositional adjustments: Conditional compositional score matrix adjustment (dropdown)
- Filters and Masking:**
 - Filter: ☐ Low complexity regions
 - Mask: ☐ Mask for lookup table only, ☐ Mask lower case letters

At the bottom, there is a 'BLAST' button and a checkbox for 'Show results in a new window'.

Figure 2.1: Algorithmic Parameters used for BLASTP in Question 1

- **Max target sequences** – Maximum number of reported hits
- **Expect threshold (E-value)** – Statistical significance cutoff value
- **Word size** – Length of the initial exact match seed used to begin alignments
- **Substitution matrix (e.g: BLOSUM62)** – Amino acid scoring scheme
- **Gap costs (Existence and Extension)** – Penalty values
- **Compositional score matrix adjustment** – Corrects amino acid composition bias

Question 3

The screenshot shows the BLAST results page for a search of '1336093|Genbank|Outer membrane integral membrane protein' against the 'nr' database. The results show a single hit from 'Pseudomonas fluorescens' (AAK81929.1) with a 97% identity and an E-value of 0.0.

Job Title: 1336093|Genbank|Outer membrane integral membrane protein
RID: TZNMKHZD114
Program: Blast 2 sequences
Query ID: Icl|Query_7705313 (amino acid)
Query Descr: 1336093|Genbank|Outer membrane integral membrane protein
Query Length: 676
Subject ID: AAK81929.1 (amino acid)
Subject Descr: RscC [Pseudomonas fluorescens]
Subject Length: 713

Filter Results:
Percent Identity: [] to []
E value: [] to []
Query Coverage: [] to []
Buttons: Filter, Reset

Clusters producing significant alignments:
Download: [v] Select columns: [v] Show: 100 [v]
[x] select all 1 clusters selected

	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
[x]	Pseudomonas fluorescens	530	530	97%	0.0	43.20%	713	AAK81929.1

Figure 3.1: Determining sequence identity of the query sequence with AAK81929.1

The "Align two or more sequences" option was selected in BLASTP, and the sequence identity and similarity of the query sequence AAK81929.1 was determined.

The percentage identity and similarity were found to be **43%** and **61%** respectively

Range 1: 22 to 690 [GenPept](#) [Graphics](#)

[▼ Next Match](#) [▲ Previous Match](#)

Score	Expect	Method	Identities	Positives	Gaps
530 bits(1366)	0.0	Compositional matrix adjust.	292/676(43%)	416/676(61%)	28/676(4%)
Query 8	RCRLLGALLMLCATLPAG--	AQTPADWKEQSYAYSADRTPLSTVLQDFADGHSVDLHLG	64		
Sbjct 22	R + +L+ C PA A PA+WK +YAY AD PL VL+DFA L +	RAKQWLVLGICIMAPAHNLLAAIPAEWKNTAYAYEADHKPLREVLEDAQTFGTQLQIE	81		
Query 65	NVEDTEVTAKIRAENASAFLDRLALEHHFQWFVYNNNTLYVSPQDEQSSERLEISPDAAPD	124			
Sbjct 82	+ + +V KIRA + LDRL +EH FQW++YNNNTL+VS D+Q S RLE+S + D	GLLEGDVNGKIRANTPQSMLEDRLGVEHFRQWYLYNNNTLFVSTLDQQESARLEVSSETISD	141		
Query 125	IKQALSGIGLLDPRFGWGLPDDGVVLVTGPPQYLELVKRFSEQREKKEDRRKVMTFPLR	184			
Sbjct 142	+KQAL+ IGLLD RFGWGLP+DGVVLV+GP Y++ +K+FS +R +++++ V++FPL+	LKQALTDIGLLDSRFGWGLPEDGVVLVSGPKTYIDQIKQFSSKRRSADEKQSVLSFPLK	201		
Query 185	YASVADRTIHYRDQTVVIPGVATMLNELMNGKRAAPASA-SGIDSTPGGPDNMMQNTQ	243			
Sbjct 202	+A+ ADR + YR + +V+PGVA +L L+ + A+ + S DS+ P T ++ +	FANAADRVKDYRGEKLVPGVANILRGLLEPRASLTGMSQPDSSQSPPLTPNVPRLG	261		
Query 244	TLLSRLSSRNKTSNRAGGRDN-----EIEDVSGRISADVRNALLIRDDDKRHDEYS	295			
Sbjct 262	LL ++ N AG D + R+ ADVRNNA+LI D +R Y	PLLQMLGAN--GNAGQLDTGPTVTPRAPVSKSRIRVEADVRRNAVLIYDLPERQAMYR	318		
Query 296	QLIAKIDVPQNLVEIDAVILDIDRTALNRLEANWQATLGGVTGGSSLSMSGSTLFVSDFK	355			
Sbjct 319	LI ++DV + L+EIDA+ILDI+RT L NW GG ++ G+ + D +	DLITQLDVARKLIEIDAIIIDIERTQLREFGVNWGFQNSRFRGGVNMAGTSSQVSIHR	378		
Query 356	-RFFADIQALEGEGTASIVANPSVLTLNQPAVIDFSQTAYITATGERVADIQPVTAGTS	414			
Sbjct 379	RF+AD+ + G+G A++V+NPSVLTLNQPAVIDF++T YI+ G A I PVT GTS	DRFYADMPSTGGQGPATMVSNPSVLTLNQPAVIDFNRTQYISP-GRDYATILPVTVGTS	437		
Query 415	LQVTPRAVGNHSSSIQLMIDIEDGHV-QTNG--DGQATGVKRGTVSTQALISENRALVL	471			
Sbjct 438	LQV PR G I L++DIEDG++ +TN D V+RG VSTQA++ E R+LV+	LQVPRVTTGRGVHQIHLVVDIEDGNLDETNPDPNHLDVRRGKVSTQAVMQEKRLV	497		
Query 472	GGFHVEESADRRRIPLLGDIPWLGQ-LFSSKRHEISQRQRLFILTPRLIGDQTDPTRYV	530			
Sbjct 498	GGFHV +S+D+ ++IPLLGDIP LG+ L SS ++R+RLFILTPR+IGDQ DP+RY+	GGFHVTDSSDQKKIPLLGDIPLLGKTLVSSTERHNNRRERLFILTPRVIGDQDDPSRYL	557		
Query 531	TADNRQQLSDAMGRVERRHSS---VNQHDVVENALRDLAEGQSPAGFQPQTSGLRLSEV	586			
Sbjct 558	D++ +L A+ + RR+S + + D++ R L G+ P F L+ +	PQDDQAEQLAALTPLARRYSPHQPVIKRSDIITTLAR-LVSGEVPKAFNAARMPLGLNTL	616		
Query 587	CRSTPALLFESTRGQWYSSSTNGVQLSVGVVNTSSKPLRFDEANCASKRTLAVAVWPHS	646			
Sbjct 617	C + L + R QWY+ V +V V+RN + +R DE C++ +TLAV VWP +	CSTRDLLALNTERSQWYAGPDYNV--AVVVLNRNQFKNVRIDEKECSNSQTLAVTVWPR	674		
Query 647	ALAPGESAEVYLAMP	662			
Sbjct 675	L PGE AEV++AM P	WLKPGEEAEVFIAMRP	690		

Figure 3.2: Sequence Alignment, with Identity and Similarity percentages shown

Question 4

Using UniProtKB, the sequences of human hemoglobin beta (**P68871**), and chicken hemoglobin beta was (**P02112**) was obtained.

These sequences were used along with the "Align two or more sequences" option in BLASTP, to obtain the results shown below:

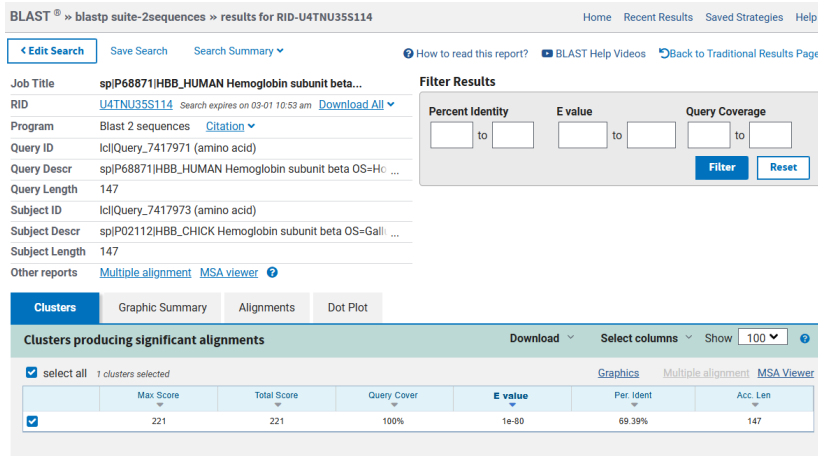


Figure 4.1: Sequence Alignment using BLASTP

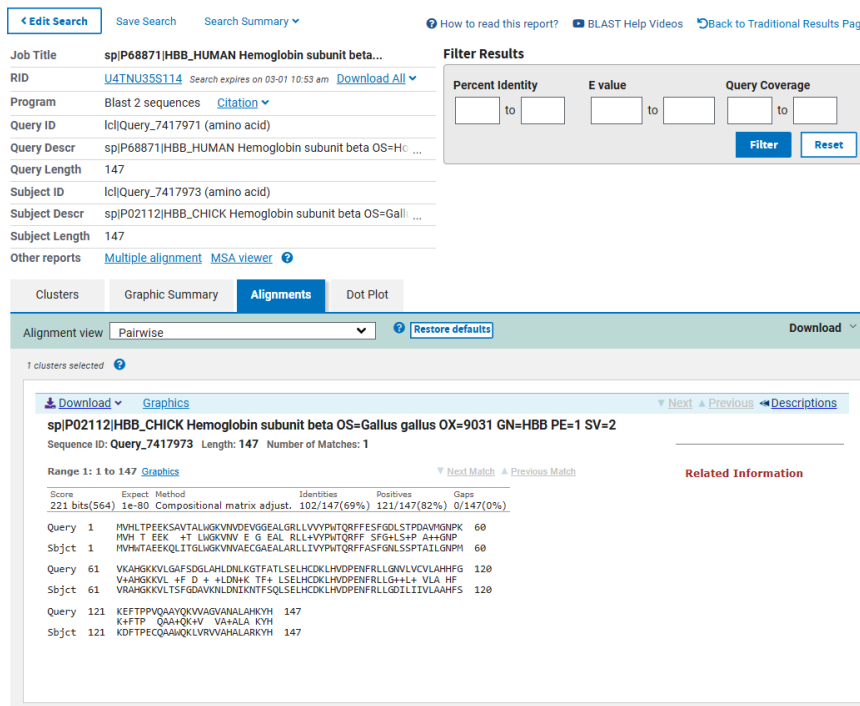


Figure 4.2: Sequence Alignment, with Identity and Similarity percentages shown

The percentage identity and similarity were found to be **69%** and **82%** respectively.

Question 5

The Python code to list all the matching pentapeptides, is given below:

```
def matching_pentapeptides(seq1,seq2):

    penta_1 = {}
    match_locations = {}

    for res in range(len(seq1)-4):

        if penta_1.get(seq1[res:res+5]) is None:
            penta_1[seq1[res:res+5]] = 1
```

```

else:
    penta_1[seq1[res:res+5]] += 1

for res in range(len(seq2)-4):

    if penta_1.get(seq2[res:res+5]) is not None:

        if match_locations.get(seq2[res:res+5]) is None:
            match_locations[seq2[res:res+5]] = (penta_1.get(seq2[res:res+5]), 1)

        else:
            match_locations[seq2[res:res+5]] = (
                match_locations.get(seq2[res:res+5])[0],
                match_locations.get(seq2[res:res+5])[1]+1
            )

    for subseq in match_locations.keys():
        print(f"The pentatpeptide {subseq} occurs {match_locations[subseq][0]} times
            in Sequence 1 and {match_locations[subseq][1]} times in Sequence 2")

seq1 = "MVHLTPEEKSAVTALWGKVVNDEVGGEALGRLLVVYPWTQRFFESFGDLSTPDVGMGNPKVKAHGKKVLGAFSD
GLAHLNDNLKGTFTATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAVQKVVAGVANALAHKYH"
seq2 = "MVHWTAEKQLITGLWGKVNVAECGAEALARLLIVYPWTQRFFASFGNLSPTAILGNPMVRAHGKKVLTSGF
DAVKNLDNIKNTFSQLSELHCDKLHVDPENFRLLGDILIIVLAHFSDFTPECQAAWQKLVRVVAHALARKYH"

matching_pentapeptides(seq1,seq2)

```

The obtained output of the program is given below:

```

The pentatpeptide LWGKV occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide WGKVN occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide GKVVN occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide VYPWT occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide YPWTQ occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide PWTQR occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide WTQRF occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide TQRFF occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide AHGKK occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide HGKKV occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide GKKVL occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide LSELH occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide SELHC occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide ELHCD occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide LHCDK occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide HCDKL occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide CDKLH occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide DKLHV occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide KLHVD occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide LHVDP occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide HVDPE occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide VDPEN occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide DPENF occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentatpeptide PENFR occurs 1 times in Sequence 1 and 1 times in Sequence 2

```

The pentapeptide ENFRL occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentapeptide NFRLL occurs 1 times in Sequence 1 and 1 times in Sequence 2
The pentapeptide FRLLG occurs 1 times in Sequence 1 and 1 times in Sequence 2

Question 6

The code to compute sequence identity, similarity, query coverage and gap percentage from the alignment sequence provided by BLASTP is given below:

```
def alignment_paramaters(align_seq, query_seq):

    identity, similarity, gaps, mismatches = 0, 0, 0, 0
    coverage = (len(align_seq)/len(query_seq)) * 100

    for res in align_seq:

        if res.isalpha():
            identity += 1
            similarity += 1

        elif res == "+":
            similarity += 1

        elif res == "-":
            gaps += 1

        else:
            mismatches += 1

    scores={}
    scores["identity"] = (identity/len(align_seq))*100
    scores["similarity"] = (similarity/len(align_seq))*100
    scores["gaps"] = (gaps/len(align_seq))*100
    scores["mismatches"] = (mismatches/len(align_seq))*100

    print(f"The query coverage is {coverage} %")
    print(f"The sequence identity is {scores["identity"]} %")
    print(f"The sequence similarity is {scores["similarity"]} %")
    print(f"The gap occurrence is {scores["gaps"]} %")

seq1="MVHLTPEEKSAVTALWGKVVNDEVGGEALGRLLVYPWTQRFFESFGDLSTPDVGMGNPKVKAHGKKVLGAFSDG
LAHLDNLKGTATLSELHCDKLHVDPENFRLLGNVLVCVLAHHFGKEFTPPVQAAVQKVVAGVANALAHKYH"

seq2="MVHWTAEKQLITGLWGKVNVAECGAEALARLLIVYPWTQRFFASFGNLSPTAILGNPMVRAHGKKVLTSFGDA
VKNLDNIKNTFSQLSELHCDKLHVDPENFRLLGDILIIVLAHFSKDFTEPCQAAWQKLVRVVAHALARKYH"

align_seq = "MVH T EEK +T LWGKVVN E G EAL RLL+VYPWTQRFF SFG+LS+P A++GNP V+AHGKKV
L +F D + +LDN+K TF+ LSELHCDKLHVDPENFRLLG++L+ VLA HF K+FTP QAA+QK+V VA+ALA KYH"

alignment_paramaters(align_seq, seq1)
```


MAFFT

MAFFT-L-INS-i Result for TIM Barrel Proteins MSA

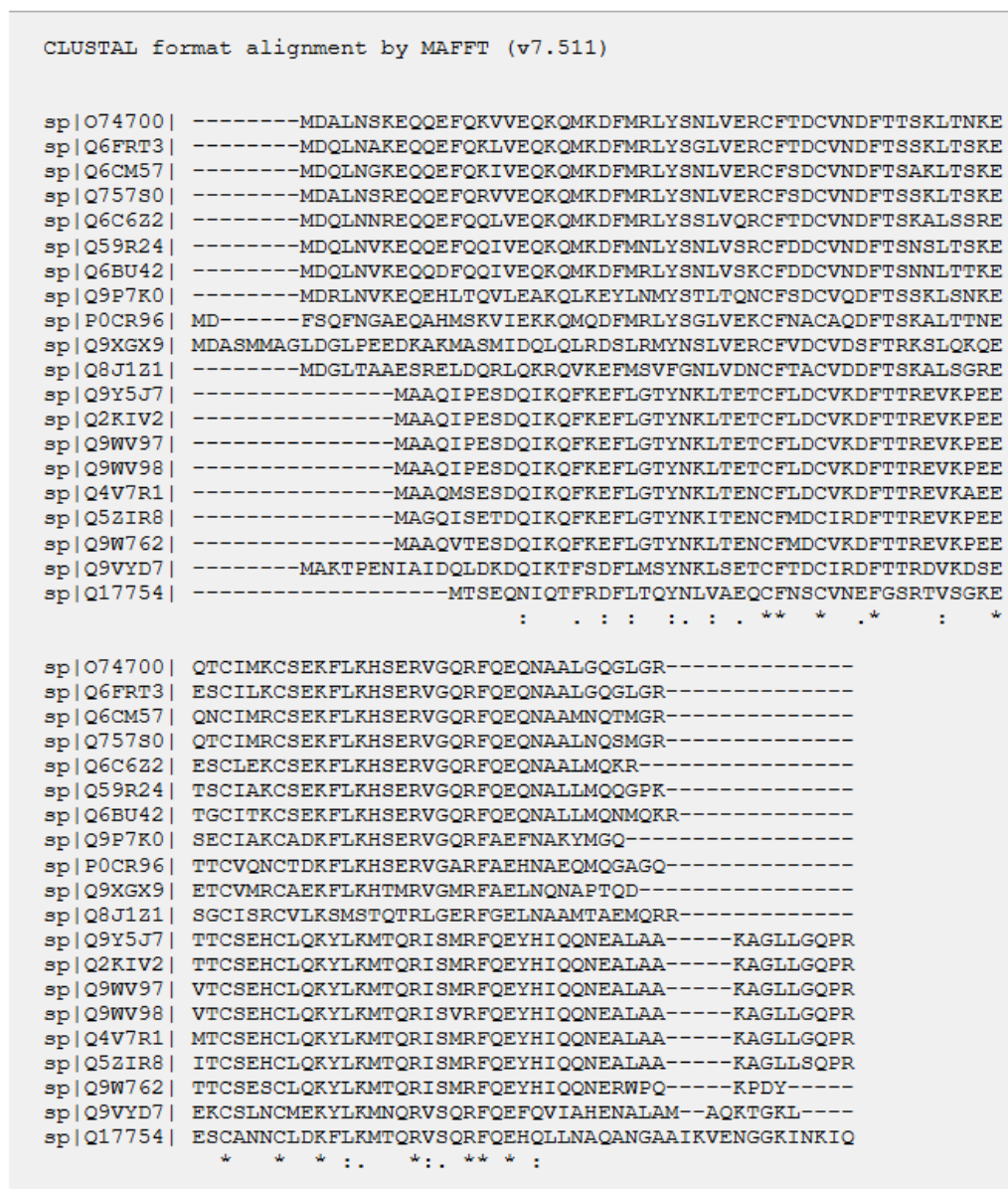


Figure 7.3: MAFFT, Multiple Sequence Alignment

MUSCLE



Figure 7.4: MUSCLE, Multiple Sequence Alignment

5 residues that were aligned differently are shown below;

Residue Postion	Clustal Omega	MAFFT	MUSCLE
1	Gap in P0CR96	M in P0CR96	Gap in P0CR96
2	Gap in P0CR96	D in P0CR96	Gap in P0CR96
8	Gap in Q17754	Gap in Q17754	M in Q17754
22	S in Q17754	S in Q17754	Gap in Q17754
100	Q in Q9W762	Gap in Q9W762	Gap in Q9W762

Question 8

BLASTP was executed on the given sequence (EPDMRTPIAHTMAW) with “pdb” as the database:

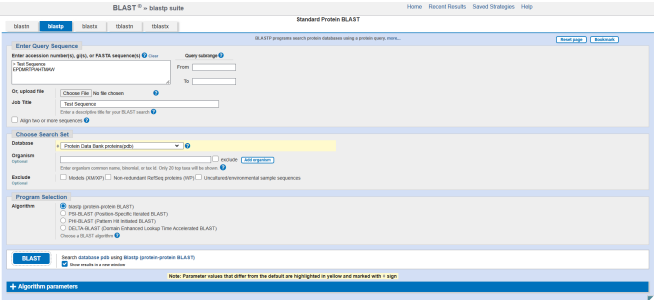


Figure 8.1: BLASTP using "pdb" database

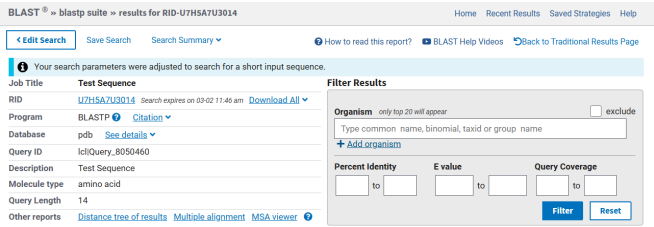


Figure 8.2: BLASTP using "pdb" database

Sequences producing significant alignments									
Download Select columns Show 100									
select all 100 sequences selected									
GenPept Graphics Distance tree of results Multiple alignment MSA Viewer									
Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli str. K-12 substr. W3110]	Escherichia coli s...	53.2	53.2	100%	2e-10	100.00%	424	2EGH_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli K-12]	Escherichia coli ...	53.2	53.2	100%	2e-10	100.00%	420	3ANL_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli K-12]	Escherichia coli ...	53.2	53.2	100%	2e-10	100.00%	410	3R0I_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli ...	53.2	53.2	100%	2e-10	100.00%	406	1Q0L_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli K-12]	Escherichia coli ...	53.2	53.2	100%	2e-10	100.00%	398	1K5H_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli ...	53.2	53.2	100%	2e-10	100.00%	398	1T1R_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli ...	43.1	43.1	100%	1e-06	85.71%	406	1Q0H_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Escherichia coli]	Escherichia coli ...	43.1	43.1	100%	1e-06	85.71%	400	1JVS_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Zymomonas mobilis]	Zymomonas mob...	42.2	42.2	93%	2e-06	84.62%	388	1R0K_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Yersinia pseudotuberculosis YPIII]	Yersinia pseudot...	40.9	40.9	86%	5e-06	91.67%	401	3JIE_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Acinetobacter baumannii AB307-0294]	Acinetobacter ba...	35.4	35.4	93%	5e-04	76.92%	406	4ZNE_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Moraxella catarrhalis]	Moraxella catarrh...	33.3	33.3	86%	0.003	75.00%	432	4ZQE_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Moraxella catarrhalis]	Moraxella catarrh...	33.3	33.3	86%	0.003	75.00%	415	4ZQG_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Vibrio vulnificus CMCP6]	Vibrio vulnificus ...	28.2	28.2	86%	0.18	75.00%	427	5KQQ_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Vibrio vulnificus CMCP6]	Vibrio vulnificus ...	28.2	28.2	86%	0.18	75.00%	405	5KRR_A	
Chain A_2-oxoglutarate-dependent ethylene/succinate-forming enzyme [Penicillium digitatum Pd1]	Arabidopsis thali...	26.1	39.0	93%	1.0	69.23%	414	8QWM_A	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Toxoplasma gondii]	Penicillium digitat...	25.2	25.2	71%	2.1	70.00%	407	9EIR_A	
Chain H_uS9m [Brassica oleracea var. botrytis]	Toxoplasma gondii...	24.8	24.8	93%	3.0	53.85%	472	8S6S_A	
Chain A_Acyl-CoA dehydrogenase [Streptomyces ficellus]	Brassica oleracea...	24.4	24.4	43%	4.2	100.00%	383	9EVS_H	
Chain A_Glutamate dehydrogenase 1 [Arabidopsis thaliana]	Streptomyces fic...	24.4	24.4	43%	4.2	100.00%	384	8HK0_A	
Chain Bh_3S5 ribosomal protein S9_mitochondrial [Arabidopsis thaliana]	Arabidopsis thali...	24.4	37.3	93%	4.2	64.29%	414	6YEH_A	
Chain A_Fusion protein of Neuronal acetylcholine receptor subunit alpha-3 and Soluble cytochrome b562 [synthetic... synthetic construct]	Arabidopsis thali...	24.4	24.4	43%	4.2	100.00%	430	6XYW_Bh	
Chain A_1-deoxy-D-xylose 5-phosphate reductoisomerase [Thermotoga maritima]	Thermotoga marit...	24.4	24.4	71%	4.2	63.64%	625	6PV7_A	
		24.0	24.0	50%	6.0	85.71%	376	3A0S_A	

Figure 8.3: List of significant alignments

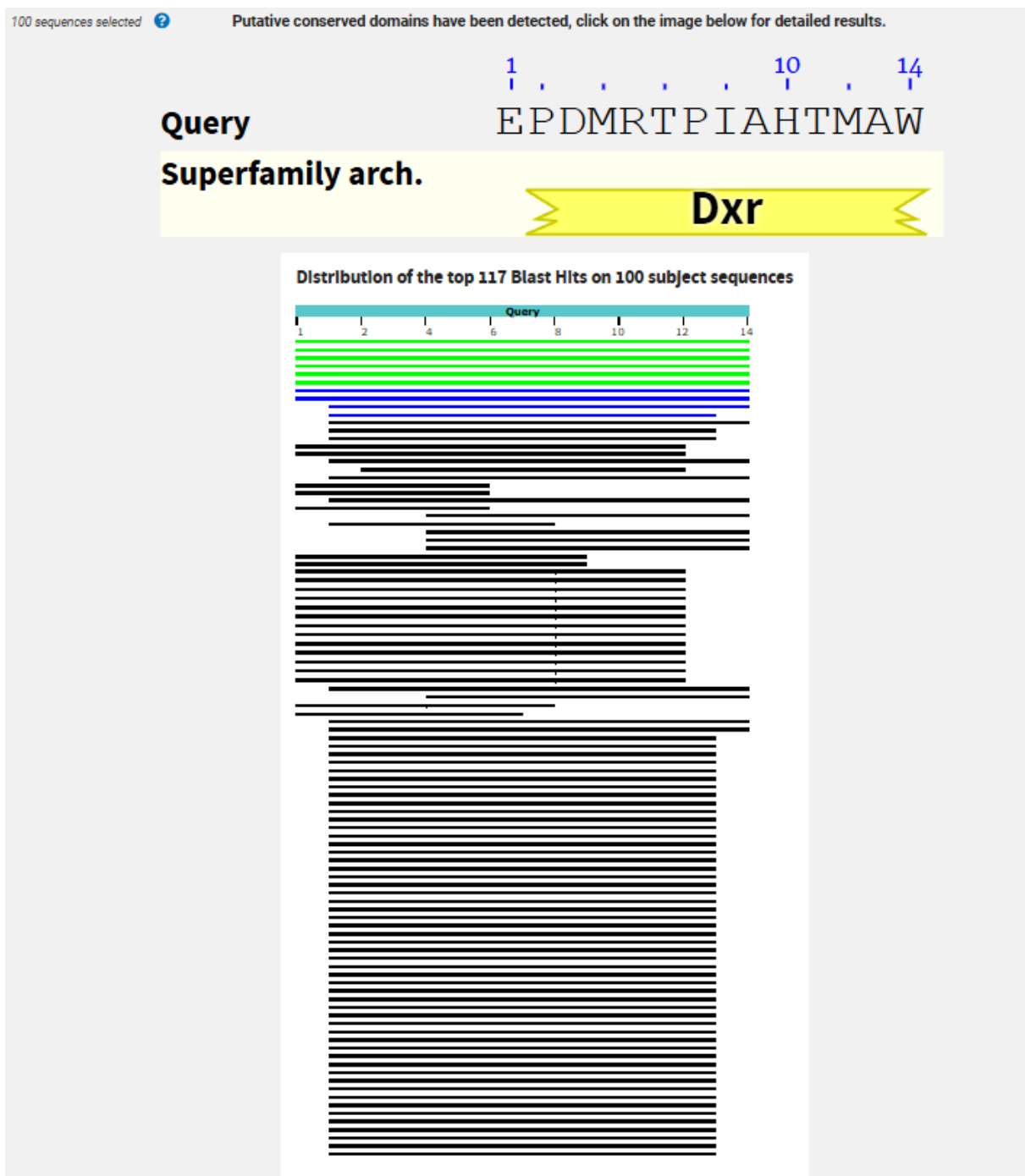


Figure 8.4: Graphic Summary of the Sequence Alignment

From Figure 8.4, we observe that the given query sequence is closely related to the Dxr superfamily. The query closely matches Chain A of 1-deoxy-D-xylulose 5 phosphate reductoisomerase, *Escherichia coli*

Superfamily - A group of proteins that share a common evolutionary origin, even if their sequences are no longer very similar (defined by their conserved 3D structural core and similar fold).

The Dxr superfamily consists of proteins related to 1-deoxy-D-xylulose-5-phosphate reductoisomerase (Dxr). It is present in most bacteria and plant chloroplasts and absent in humans, making it a perfect target for antibiotic and antimalarial drugs.

Additionally since our query sequence is quite short, many of the results have 100% query coverage, and E values very close to 0.

Link to the google drive containing the python code used in this assignment: https://drive.google.com/drive/u/0/folders/1ZkaZCU_6aC0arfT5X6bsZ_gBcHoD0P2h

Annotations have been omitted from the submitted PDF for the sake of brevity. The fully annotated version of the code is available in the code folder within the attached Google Drive.
